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orbitize! v3: Orbit-fitting for the High-contrast Imaging Community

Sarah Blunt[1,2¶], **Jason Jinfei Wang**[1], **Lea Hirsch**[11], **Roberto Tejada**[12], **Vighnesh Nagpal**[13], **Tirth Dharmesh Surti**[5], **Sofia Covarrubias**[2], **Thea McKenna**[1,6,7,8], **Rodrigo Ferrer Chávez**[1], **Jorge Llop-Sayson**[4], **Mireya Arora**[14], **Amanda Chavez**[1], **Devin Cody**[9], **Saanika Choudhary**[1], **Adam Smith**[15], **William Balmer**[16], **Tomas Stolker**[3], **Hannah Gallamore**[15], **Clarissa R. Do Ó**[10], **Eric Nielsen**[15], and **Robert J. De Rosa**[17]

1 Center for Interdisciplinary Exploration and Research in Astrophysics (CIERA), Northwestern University **2** California Institute of Technology **3** Leiden Observatory, Leiden University **4** Jet Propulsion Laboratory, California Institute of Technology **5** Department of Physics/Kavli Institute for Particle Astrophysics and Cosmology, Stanford University **6** Southeastern Universities Research Association **7** Astronomy Department, Cornell University **8** NASA Goddard Space Flight Center, Code 698 **9** Google, Inc. **10** Department of Astronomy & Astrophysics, University of California San Diego **11** Department of Chemical & Physical Sciences, University of Toronto Mississauga **12** Department of Astrophysical Sciences, Princeton University **13** University of California, Berkeley **14** University of Michigan **15** New Mexico State University **16** Johns Hopkins University **17** European Southern Observatory ¶ Corresponding author

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Software

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- @DTCupcakes

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Summary

`orbitize!` is a package for Bayesian modeling of the orbital parameters of resolved binary objects from timeseries measurements. It was developed with the needs of the high-contrast imaging community in mind, and has since also become widely used in the binary star community. A generic `orbitize!` use case involves translating relative astrometric timeseries, optionally combined with radial velocity or astrometric timeseries, into a set of derived orbital posteriors.

This paper is published alongside the release of `orbitize!` version 3.0, which has seen significant expansions in functionality and accessibility since the release of version 1.0 (Blunt et al., 2020).

Statement of need

The orbital parameters of directly-imaged planets and binary stars can tell us about their present-day dynamics and formation histories (Bowler, 2016), as well as about their inherent physical characteristics (particularly mass, generally called "dynamical mass" when derived from orbital constraints, e.g. (Brandt, 2021), (Lacour et al., 2021)).

`orbitize!` is used widely in the exoplanet imaging and binary star communities for translating astrometric data to information about eccentricities (Bowler et al., 2020), obliquities (Bryan et al., 2020), dynamical masses (Lacour et al., 2021), and more.

Each new released version of the `orbitize!` source code is automatically archived on Zenodo (Blunt et al., 2023).

Major features added since v1

For a detailed overview of the `orbitize!` API, core functionality (including information about our Kepler solver), and initial verification, we refer readers to (Blunt et al., 2020). This section lists major new features that have been added to the code since the release of version 1.0 and directs readers to more information about each. A complete descriptive list of modifications to the code is maintained in our changelog.

Major new features since v1 include:

1. The ability to jointly fit radial velocity (RV) timeseries, both RVs of the secondary companion (see Section 3 of [Blunt:2023a]) and RVs of the primary star. RVs of the primary star can either be passed into `orbitize!` directly (see the radial velocity tutorial), or fit separately and passed in as prior information (see the non-orbitize! posteriors as priors tutorial.)

2. The ability to jointly fit absolute astrometry of the primary star. `orbitize!` can fit the Hipparcos-Gaia catalog of accelerations (Brandt, 2021) (see the HGCA tutorial), as well as Hipparcos intermediate astrometric data and Gaia astrometry, following (Nielsen et al., 2020) (see the Hipparcos IAD tutorial). It can also handle arbitrary absolute astrometry (Sarah to add tutorial link).

3. In addition to the MCMC and OFTI posterior computation algorithms documented in (Blunt et al., 2020), `orbitize!` version 3 also implements a nested sampling backend, via `dynesty` (Speagle, 2020) (see the `dynesty` tutorial.)

4. `orbitize!` version 3 implements two prescriptions for handling multi-planet effects. Keplerian epicyclic motion of the primary star due to multiple orbiting bodies, following (Lacour et al., 2021), is discussed in the multi-planet tutorial, and N-body interactions are discussed in (Covarrubias et al., 2022). The Keplerian epicyclic motion prescription only accounts for star-planet interactions, treating the motion of the star as a sum of Keplerians, while the N-body prescription models this effect as well as planet-planet interactions.

5. The ability to fit in different orbital bases (Surti et al., 2023), (Ferrer-Chávez et al., 2021) (see the changing basis tutorial), as well as the ability to apply the observation-based priors derived in (O'Neil et al., 2019) (see the observation-based priors tutorial).

Verification and Documentation

`orbitize!` implements a full stack of automated testing and documentation building practices. We use GitHub Actions to automatically run a suite of unit tests, maintained in orbitize/tests, each time code is committed to the public repository or a pull request is opened. The jupyter notebook tutorials, maintained in orbitize/docs/tutorials, are also automatically run when a pull request to the `main` branch is opened. Documentation is built using `sphinx`, and hosted on readthedocs.org at orbitize.info. We also maintain a set of longer-running tests in orbitize/tests/end-to-end-tests that show real scientific use cases of the code. These tests are not automatically run.

`orbitize!` is documented through API docstrings describing individual functions, which are accessible on our readthedocs page, a set of jupyter notebook tutorials walking the user through a particular application, a set of frequently asked questions, and an in-progress "manual" describing orbit-fitting with `orbitize!` from first principles.

DRAFT

Comparison to Similar Packages

Since the release of `orbitize!` version 1, other open-source packages have been released that have similar goals to `orbitize!`, notably `orvara` (Brandt, 2021) and `octofitter` (Thompson et al., 2023). This is a wonderful development, since all packages benefit from open sharing of knowledge! `orbitize!`, `orvara`, and `octofitter` can do many similar things, but each has unique features and strengths; as an example, `octofitter` is extraordinarily fast, and enables joint astrometry extraction and orbit modeling, while `orbitize!` has unique abilities to fit arbitrary absolute astrometry (i.e. not from Hipparcos or Gaia) and model data using an N-body backend. `orvara` analytically marginalizes over parallax assuming a prior informed by Gaia, a significant speed advantage, while `orbitize!` allows different parallax priors to be used. We recommend users of each package compare the implementations of the particular features they wish to use.

Best practices for orbit-fitting, particularly using radial velocities, for which treatment of stellar activity is an active area of research, and absolute astrometry with Gaia and Hipparcos, for which correlated error treatment is an active area of research, evolve quickly. The philosophy of `orbitize!` is to, as much as possible, implement multiple approaches to a problem, evidenced by our multiple implementations of radial velocity joint fitting and absolute astrometry joint fitting (described above). For detailed information about our particular implementations, we direct the reader to our documentation.

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DRAFT

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